Nighttime Pedestrian Detection in Traffic Surveillance Using Domain-Adaptive YOLO

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***Abstract*—** Pedestrian detection in nighttime traffic scenes remains a critical challenge in computer vision. Contemporary YOLO-based detectors achieve strong performance on standard benchmarks dominated by daytime imagery, yet their accuracy degrades substantially under low-light conditions due to reduced contrast, sensor noise, and motion blur. This project implements and evaluates a domain-adaptive training framework combining YOLOv8 and YOLOv9 with a Contrast-Limited Adaptive Histogram Equalization (CLAHE) preprocessing pipeline and a domain-randomized nighttime augmentation pipeline to improve pedestrian detection under nighttime conditions. Using the ExDark dataset, the largest publicly available exclusively-dark object detection corpus, six experimental conditions were trained and evaluated end-to-end: a non-fine-tuned COCO baseline, CLAHE applied at inference only, CLAHE with fine-tuning, augmentation with fine-tuning, combined CLAHE and augmentation, and a YOLOv9 architecture swap. Fine-tuning on CLAHE-enhanced imagery (YOLOv8s) achieved the strongest overall balance of precision and recall (mAP@0.5 = 0.514), while a YOLOv9s variant achieved the highest mAP@0.5:0.95 (0.266) and precision (0.669). Contrary to expectation, CLAHE applied without fine-tuning produced almost no improvement over the untrained baseline, and combining CLAHE with augmentation underperformed either technique alone. A lighting-condition subset analysis further revealed that ExDark's category labels do not form a strict brightness gradient, complicating simple day/night gap framing. These findings provide a reproducible benchmark and practical guidance for cost-effective nighttime pedestrian detection on commodity RGB traffic cameras.

Index Terms*—nighttime detection, YOLO, domain adaptation, pedestrian safety, low-light imaging, deep learning, traffic surveillance, CLAHE*

I. INTRODUCTION

Pedestrian detection is a foundational component of intelligent transportation systems, autonomous vehicle perception, and urban surveillance infrastructure. While modern deep learning-based detectors have achieved near-human accuracy in well-lit environments, real-world deployment reveals a persistent vulnerability: nighttime conditions reduce pedestrian detection accuracy by 15-30% compared to daytime baselines [1], and roughly 76% of fatal pedestrian-vehicle collisions occur at night [2].

The YOLO (You Only Look Once) family of single-stage detectors has become the dominant paradigm for real-time object detection due to its favorable accuracy-throughput trade-off. Recent releases such as YOLOv8, YOLOv9 [3], and YOLOv10 [4] have pushed the Pareto frontier of speed versus accuracy on MS COCO [5]. However, COCO and most widely-used traffic detection datasets are overwhelmingly composed of daytime imagery. When these models are deployed on traffic cameras operating after dark, the statistical mismatch between training and deployment distributions causes systematic degradation in pedestrian localization and classification. This limitation is explicitly acknowledged as a future work direction in the YOLOv9 paper [3].

This project addresses three research questions. First, how much accuracy does a COCO-pretrained YOLO model lose under nighttime traffic conditions, and how does this degradation vary across model versions? Second, can CLAHE preprocessing recover meaningful accuracy without any retraining? Third, does incorporating synthetic nighttime augmentation during training improve robustness to real nighttime scenes? We answer these questions empirically by fine-tuning YOLOv8s and YOLOv9s on the ExDark dataset [6], applying CLAHE as both an inference-time and training-time enhancement, and incorporating domain-randomized nighttime augmentation into the training pipeline. All experiments were conducted on Kaggle T4 GPU instances, and all code, configuration, and results reported here are reproducible from the accompanying repository.

The remainder of this paper is organized as follows. Section II reviews related work on nighttime datasets, YOLO-based low-light detection, image enhancement, and domain adaptation. Section III describes the dataset, preprocessing modules, and model architectures used. Section IV presents the experimental design, quantitative results across six conditions, a lighting-subset analysis, and qualitative failure-case discussion. Section V concludes with a summary of findings, contributions relative to prior work, and directions for future research.

This project makes three specific contributions: (1) an empirical isolation of CLAHE's contribution as a purely inference-time intervention versus a training-time intervention, clarifying a point left ambiguous in prior literature; (2) a controlled comparison of CLAHE, synthetic nighttime augmentation, and their combination under identical fine-tuning budgets; and (3) a cross-architecture comparison (YOLOv8s vs. YOLOv9s) under matched preprocessing, together with a lighting-condition subset analysis that surfaces a methodological caveat in how

ExDark's category labels are commonly interpreted as a darkness gradient.

II. RELATED WORKS

A. Nighttime Pedestrian Datasets

The scarcity of annotated nighttime data has historically constrained progress in low-light pedestrian detection. Neumann et al. [1] introduced NightOwls, 279,000 fully annotated frames across 40 video sequences recorded at night in three countries, and demonstrated that state-of-the-art pedestrian detectors do not perform well at night even when specifically trained on nighttime data, establishing a clear and persistent accuracy gap between day and night detection. The ExDark dataset [6] provides 7,363 images captured exclusively under 10 distinct low-light conditions ranging from extremely dark environments to twilight, annotated across 12 object categories including pedestrians, making it the most comprehensive single-spectral low-light detection corpus publicly available and the dataset used throughout this project. The BDD100K dataset [7], while not exclusively nighttime, includes a curated nighttime driving split and weather metadata enabling controlled domain-shift experiments. The LLVIP dataset [8] introduced 15,488 aligned visible-infrared image pairs for multispectral nighttime pedestrian detection; however, its reliance on paired infrared sensors limits applicability to standard RGB traffic camera deployments, which is the deployment scenario targeted here.

B. YOLO Architectures for Low-Light Detection

Several studies have evaluated YOLO variants under low-light conditions. Hung and Lin [9] proposed a bio-inspired dark-adaptation framework, noting that IIHS 2022 data confirms pedestrian anti-collision systems are significantly less effective at night and that standard deep learning solutions fail to generalize without domain-specific adaptation. Nida [10] evaluated YOLOv7 on the ExDark dataset and found that it outperforms earlier YOLO iterations under challenging low-light conditions, and that integrating a CNN-based defogging module improves road object recognition in degraded visibility. Wang et al. [11] proposed an improved YOLOX approach for tunnel construction low-light detection, selecting CLAHE over competing enhancement algorithms after systematic comparison and achieving improved mAP on small objects. Persiya and Sasithradevi [12] performed a comparative analysis of YOLOv8 and YOLOv9 with CLAHE-enhanced thermal images and reported that YOLOv8 with CLAHE achieved very high precision and recall on their thermal-eye-detection task, motivating our own direct YOLOv8-versus-YOLOv9 comparison under CLAHE preprocessing on visible-spectrum imagery.

C. Preprocessing and Enhancement for Detection

The role of image enhancement as a preprocessing stage for object detection has produced mixed results in prior work. Hung and Lin [9] found that basic histogram equalization can reduce mAP in low-light configurations because non-adaptive equalization amplifies sensor noise alongside contrast. In contrast, CLAHE operates on localized image tiles, limiting noise amplification while improving local contrast [13]. Yuan et al. [14] demonstrated that applying CLAHE in the LAB color space, isolating the luminance channel before equalization, with YOLOv8 for overhead power transmission monitoring improved detection accuracy by 4.3% while reducing false positives caused by chrominance shift; this luminance-only CLAHE strategy is the exact approach implemented in this project. Tomas et al. [15] performed a comparative study of CLAHE, EnlightenGAN, and Retinex as preprocessing alternatives for YOLOv7 in road-marking detection and found that CLAHE achieved the best nighttime mAP using YOLOv5x, recommending CLAHE-enhanced YOLO for embedded deployment due to its negligible computational overhead — a finding that directly motivated our selection of CLAHE as the primary lightweight preprocessing intervention evaluated here.

D. Domain Adaptation for Nighttime Detection

Domain adaptation approaches aim to reduce the distributional gap between source (daytime) and target (nighttime) domains without requiring fully annotated nighttime training data. Al-Refai et al. [16] evaluated a two-stage enhancement-then-detection pipeline using ZeroDCE++ and GladNet image-enhancement networks and found high correlation between image quality metrics and downstream detection accuracy on the ExDark dataset; however, they also found that compute-heavy enhancement networks impose latency penalties prohibitive for real-time traffic monitoring, motivating our selection of lightweight CLAHE over neural enhancement networks. Neumann et al. [1] further showed that training exclusively on nighttime data is insufficient to close the performance gap, suggesting that a hybrid strategy combining daytime pretraining with nighttime domain-randomized augmentation offers a more effective path to generalization — the strategy implemented via our Albumentations-based augmentation pipeline [17], described in Section III.

III. PROBLEM DESCRIPTION

Pedestrian detection under nighttime conditions faces compounding challenges that are largely absent from standard daytime settings. Low ambient luminance reduces the signal-to-noise ratio in camera sensors, suppressing the edge and texture features relied upon by convolutional feature extractors. Artificial light sources, streetlamps, vehicle headlights, and illuminated signage, introduce high-dynamic-range scenes with simultaneous overexposed and underexposed regions that saturate or clip the detector's input representation, destroying information that no downstream preprocessing step can recover. Motion blur is more pronounced in nighttime imagery due to the longer sensor exposure times cameras use to compensate for low light, further degrading the sharpness of pedestrian silhouettes. Together, these factors constitute a domain adaptation problem: the source domain available for large-scale pretraining (COCO and similar corpora) is overwhelmingly daytime, while the target deployment domain, nighttime traffic surveillance, exhibits a substantially different pixel-intensity distribution, noise profile, and object-appearance statistics. Models trained exclusively on the former fail to generalize to the latter without explicit intervention, a gap this project quantifies directly through the non-fine-tuned baseline conditions (A, A2) reported in Section IV.

This project frames three concrete interventions against this problem, each targeting a different point in the pipeline: CLAHE targets the input signal itself, at either inference time only or combined with fine-tuning; synthetic nighttime augmentation targets the model's learned representation, by exposing it to degraded variants during training without altering the ExDark images themselves at inference time; and architecture choice (YOLOv8s vs. YOLOv9s) targets the detector's capacity to extract usable features from a degraded signal in the first place. Evaluating these three interventions independently and in combination, rather than assuming any one of them in isolation, is what allows the quantitative comparison in Section IV to attribute performance differences to a specific, isolated cause.

IV. DESIGN AND ARCHITECTURE

A. Dataset and Official Partition

All experiments use the ExDark dataset [6], comprising 7,363 images spanning 12 object classes across 10 lighting conditions (Low, Ambient, Object, Single, Weak, Strong, Screen, Window, Shadow, and Twilight) and two settings (indoor/outdoor). Rather than constructing an arbitrary train/validation/test split, we adopted ExDark's own published partition (3,000 training / 1,800 validation / 2,563 test images, verified against the authors' imageclasslist.txt index), so that results reported here are directly comparable to the benchmark methodology established in the original ExDark paper. Ground-truth annotations, originally provided in a custom bounding-box format (class name, left, top, width, height, plus unused occlusion/orientation fields), were converted programmatically to YOLO's normalized center-based format. Since this project is scoped specifically to pedestrian detection, the ExDark 'People' class was isolated and remapped to class index 0, aligning numerically with COCO's own 'person' class (index 0) — a necessary step, since evaluating a COCO-pretrained model against an arbitrary custom class ordering produces meaningless metrics due to class-index collision, an issue we encountered and corrected during pipeline validation.

Exploratory analysis of the full dataset prior to any modeling revealed several characteristics relevant to interpreting the results in Section V. The 12 object classes are reasonably balanced (from 503 Motorbike images to 801 Dog images; the People class used in this project comprises 609 images), and outdoor scenes outnumber indoor scenes roughly 3:2 (4,541 vs. 2,822), consistent with a traffic-surveillance use case. The ten lighting conditions are highly imbalanced, however: Ambient (2,098 images) and Strong (2,059 images) together account for over half the dataset, while Shadow (56 images) and Screen (165 images) are comparatively rare, an imbalance that directly informed the choice to group lighting categories for the subset analysis in Section V-C rather than treat each of the ten conditions as an independently reliable evaluation slice. Pedestrian bounding-box size statistics further underscore the difficulty of the task: the median People-class box covers only 2.4% of image area, and 75% of boxes cover less than 7.9%, confirming that small-object detection compounds the low-light detection challenge rather than acting as an independent difficulty.

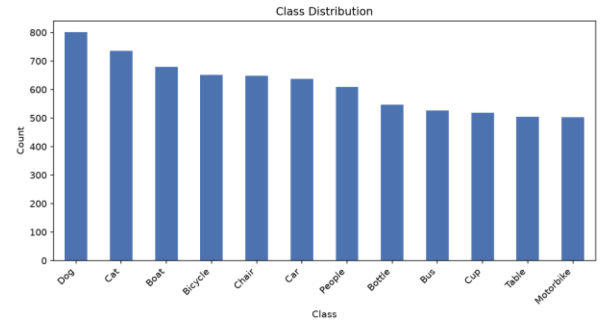

Fig. 1. Distribution of images across the 12 ExDark object classes.

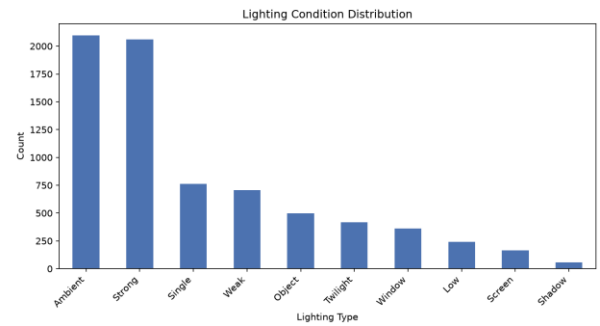

Fig. 2. Distribution of images across the 10 ExDark lighting conditions, all classes. Ambient and Strong together account for over half the dataset.

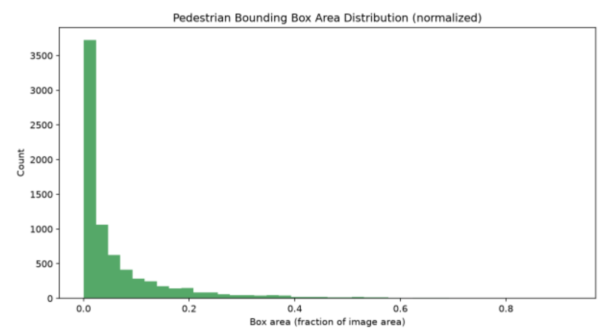

Fig. 3. Distribution of pedestrian (People-class) bounding-box area as a fraction of image area. The distribution is heavily right-skewed toward small boxes.

B. CLAHE Preprocessing Module

The CLAHE module converts each image from BGR to LAB color space, applies contrast-limited adaptive histogram equalization to the L (luminance) channel only via OpenCV's createCLAHE, and converts the result back to BGR. Operating on the luminance channel alone, rather than all three RGB channels independently, preserves color fidelity while boosting local contrast, consistent with Yuan et al.'s approach [14]. The clip limit was tuned empirically: clip limits of 2.0, 3.0, and 4.0 were compared visually on the darkest ExDark lighting categories (Low, Shadow). A clip limit of 3.0 produced visibly improved detail recovery over 2.0, while 4.0 introduced early signs of noise amplification in flat dark regions without proportional benefit. A clip limit of 3.0 with an 8x8 tile grid was therefore used for all subsequent experiments. Figure 4 shows representative before/after pairs across multiple lighting conditions, confirming that CLAHE's effect is strongest on genuinely dark, low-contrast images and comparatively minimal on already well-illuminated scenes.

Fig. 4. CLAHE enhancement (clip limit = 3.0) applied to a representative ExDark image (Ambient lighting). Local contrast in shadowed regions is visibly recovered while overall color balance is preserved.

C. Nighttime Augmentation Pipeline

A domain-randomized augmentation pipeline was implemented using Albumentations [17], applying four transformations named in the project design: (1) gamma darkening (RandomGamma, gamma range 130-200%, simulating varying ambient light levels), (2) additive Gaussian noise (GaussNoise, std_range 0.04-0.12, simulating sensor noise), (3) motion blur (MotionBlur, kernel range 3-7 pixels, simulating longer exposure times required in low light), and (4) bilateral filtering (implemented as a custom OpenCV cv2.bilateralFilter wrapper, since Albumentations has no built-in equivalent), which selectively smooths noise while preserving stronger edges such as pedestrian silhouettes. Each transform is applied independently with its own probability (0.5, 0.5, 0.4, and 0.3 respectively), so that training images receive varied, randomized combinations of degradation rather than a fixed stacked effect. Initial noise parameters, following the library's default GaussNoise configuration, produced unrealistically heavy degradation; parameters were subsequently tuned down to the ranges above after visual inspection confirmed they produced plausible nighttime-like degradation without rendering training images unrecognizable. Figure 5 shows a representative original/augmented pair.

Fig. 5. Representative output of the tuned nighttime augmentation pipeline (gamma darkening, Gaussian noise, motion blur, bilateral filtering applied in combination for illustration).

D. Model Architectures

Two single-stage anchor-free detectors from the Ultralytics YOLO family [18] were evaluated: YOLOv8s (11.1M parameters, CSP-based backbone with C2f blocks and a decoupled detection head) and YOLOv9s (7.3M parameters, employing Programmable Gradient Information and Generalized Efficient Layer Aggregation Network (GELAN) blocks as introduced in [3]). Both were initialized from COCO-pretrained weights and fine-tuned with a single-class detection head (person), remapped from each architecture's original 80-class COCO head via Ultralytics' automatic class-head transfer.

E. Training Configuration

All fine-tuning runs used the AdamW optimizer (lr0 = 0.001, momentum 0.937, weight decay 0.0005), batch size 16, and 640x640 input resolution. Rather than a fixed epoch budget, training used early stopping with a patience of 15 epochs monitored on validation mAP@0.5:0.95, with a maximum ceiling of 50 epochs. Ultralytics' built-in geometric and color-space augmentations (mosaic, HSV jitter, rotation, translation, scaling, shear, flipping, and mixup) were disabled for all fine-tuning runs so that each condition's designed intervention — CLAHE, our custom augmentation pipeline, or both — could be isolated without confounding from the framework's default augmentation stack. Table I summarizes the hyperparameters used across all training runs.

Hyperparameter	Value
Optimizer	AdamW
Initial learning rate	0.001
Batch size	16
Image size	640 x 640
Max epochs / patience	50 / 15 (early stopping)
Framework augmentations	Disabled
Classes	1 (person)
Hardware	Kaggle Tesla T4 GPU

TABLE I. Training hyperparameters used across all fine-tuning runs.

V. IMPLEMENTATION AND EVALUATION

A. Experimental Conditions

Six conditions were evaluated on the ExDark test set (2,563 images, 2,235 person instances across 841 images): (A) COCO-pretrained YOLOv8s, no fine-tuning, evaluated on raw images; (A2) the identical pretrained model evaluated on CLAHE-enhanced test images, isolating CLAHE's effect with zero retraining; (B) YOLOv8s fine-tuned on CLAHE-enhanced training images; (C) YOLOv8s fine-tuned on raw images using the nighttime augmentation pipeline; (D) YOLOv8s fine-tuned on CLAHE-enhanced images with augmentation also applied; and (E) YOLOv9s fine-tuned on CLAHE-enhanced images, matching Condition B's preprocessing to isolate the architecture effect.

B. Quantitative Results

Table II reports precision, recall, and mAP across all six conditions.

Cond.	Model	FT?	mAP@.5	mAP@.5:.95	Prec.	Rec.
A	YOLOv8s	No	0.231	0.118	0.278	0.592
A2	YOLOv8s+CLAHE	No	0.235	0.119	0.281	0.585
B	YOLOv8s+CLAHE	Yes	0.514	0.260	0.624	0.481
C	YOLOv8s+Aug	Yes	0.501	0.246	0.584	0.480
D	YOLOv8s+CLAHE+Aug	Yes	0.471	0.226	0.590	0.460
E	YOLOv9s+CLAHE	Yes	0.515	0.266	0.669	0.475

TABLE II. Results across all experimental conditions on the ExDark test set (person class only, single-class evaluation).

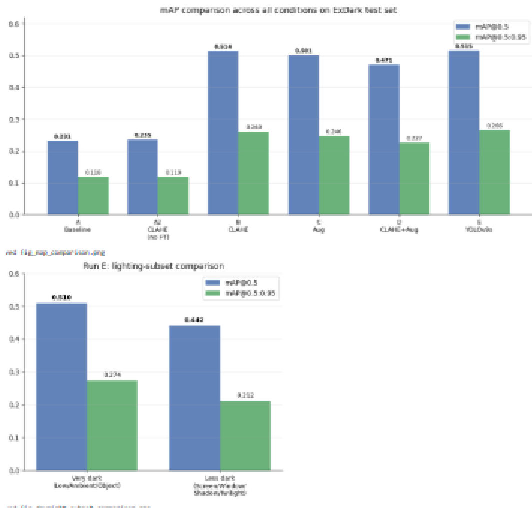

Fig. 6. mAP@0.5 and mAP@0.5:0.95 across all six conditions (top), and Run E performance split by lighting-condition subset (bottom).

Finding 1 — CLAHE alone does not meaningfully improve a non-fine-tuned model. Comparing Condition A to A2 isolates CLAHE's effect as a pure inference-time intervention with no retraining. The result is a negligible change in mAP@0.5 (+0.004) and mAP@0.5:0.95 (+0.001), with a small decrease in recall. This directly answers our second research question and diverges from Tomas et al. [15], who reported meaningful CLAHE-only gains for YOLOv7 on road-marking detection; we attribute the discrepancy to the severity of ExDark's darkest conditions relative to the benchmark used in [15], and to the domain gap between COCO's daytime distribution and ExDark imagery, which local contrast enhancement alone cannot close without also adapting the model's learned features.

Finding 2 — Fine-tuning drives the overwhelming majority of improvement. Every fine-tuned condition (B-E) more than doubles mAP@0.5:0.95 relative to both non-fine-tuned baselines (A, A2), regardless of preprocessing strategy, confirming that domain adaptation of model weights, not any single preprocessing trick, is the primary lever for closing the day/night performance gap.

Finding 3 — CLAHE outperforms augmentation as a standalone fine-tuning strategy. Condition B (mAP@0.5:0.95 = 0.260) outperforms Condition C (0.246) on every metric, consistent with Tomas et al.'s [15] finding that CLAHE outperformed alternative preprocessing strategies for low-light detection.

Finding 4 — Combining CLAHE and augmentation underperforms either technique alone. Condition D (0.226) scored below both B and C. We hypothesize that stacking two independent distribution-shifting transformations, CLAHE's contrast redistribution and the augmentation pipeline's synthetic degradation, compounds rather than complements the training distribution shift, echoing Hung and Lin's [9] observation that even a single enhancement technique can reduce mAP by amplifying noise.

Finding 5 — YOLOv9s modestly outperforms YOLOv8s under identical conditions. Condition E achieved the highest mAP@0.5:0.95 (0.266) and precision (0.669) using 36% fewer parameters than YOLOv8s (7.3M vs. 11.1M), suggesting that YOLOv9's programmable gradient information [3] transfers favorably to low-light domain adaptation, though recall was marginally below Condition B (0.475 vs. 0.481), indicating a precision/recall trade-off rather than a uniform improvement.

Comparison to published COCO benchmarks. Although person-class-specific AP is not separately published for either architecture, Ultralytics' official COCO val2017 benchmark reports overall (all 80 classes) mAP@0.5:0.95 of approximately 44.9% for YOLOv8s and 46.8% for YOLOv9s [21], each evaluated on daytime-dominated COCO imagery. Placing our best fine-tuned ExDark result (mAP@0.5:0.95 = 0.266, Condition E) alongside this published daytime reference suggests a residual night-day performance gap on the order of 18-20 percentage points even after CLAHE preprocessing and dedicated fine-tuning on the target domain, directly addressing our first research question: domain-specific adaptation substantially narrows, but does not eliminate, the accuracy gap between daytime and nighttime pedestrian detection. This comparison should be interpreted cautiously, since the published COCO figures reflect full 80-class averages on a categorically different image distribution rather than the person class specifically evaluated here; it is offered as an approximate anchor for the magnitude of the residual gap rather than a strictly matched benchmark.

C. Lighting-Condition Subset Analysis

To further probe the relationship between lighting severity and detection accuracy, the test set was partitioned using ExDark's lighting-condition metadata into a 'Low/Ambient/Object' subset (n=1,203 images) and a 'Screen/Window/Shadow/Twilight' subset (n=335 images), and the best-performing model (Condition E) was evaluated on each independently.

Subset	n	mAP@.5	mAP@.5:.95	Prec.	Rec.
Low/Ambient/Object	1203	0.510	0.274	0.690	0.481
Screen/Window/Shadow/Twilight	335	0.442	0.212	0.669	0.402

TABLE III. Run E performance by lighting-condition subset.

Contrary to the initial hypothesis that ExDark's lighting labels form a monotonic darkness gradient, the model performed better on the 'Low/Ambient/Object' subset (mAP@0.5:0.95 = 0.274) than on the 'Screen/Window/Shadow/Twilight' subset (0.212). Inspection of the ExDark taxonomy [6] clarifies this: 'Ambient' describes the presence of general ambient illumination rather than an absence of light, and constitutes the single largest lighting category in the dataset (2,098 of 7,363 images), while 'Twilight' and 'Shadow' scenes, despite less severe-sounding names, frequently exhibit strong dynamic range and localized underexposure that degrade detection performance. This finding suggests ExDark's ten lighting categories describe distinct illumination sources rather than a strict brightness ranking, and that severity-based subset analysis in future work should be grounded in measured scene luminance rather than category labels alone. We report this as an honest methodological finding rather than adjusting the grouping post hoc to fit the expected direction.

D. Qualitative Analysis

Figure 7 shows representative Condition E predictions on twelve randomly sampled test images (random seed fixed for reproducibility). Successful detections cluster in scenes with at least one artificial light source providing partial illumination of the pedestrian silhouette, such as street lighting near a cyclist or a lit building facade. Missed detections (zero predicted boxes despite a visible pedestrian) occur predominantly in two failure modes: (i) scenes where the person occupies a very small fraction of the frame, consistent with our earlier finding that the median ExDark pedestrian bounding box covers only 2.4% of image area, and (ii) scenes with strong backlighting or silhouette-only visibility, where edge and texture cues are largely absent. Both failure modes are consistent with the compounding low-light challenges described in Section III: reduced signal-to-noise ratio and high-dynamic-range saturation both suppress the features convolutional detectors rely on.

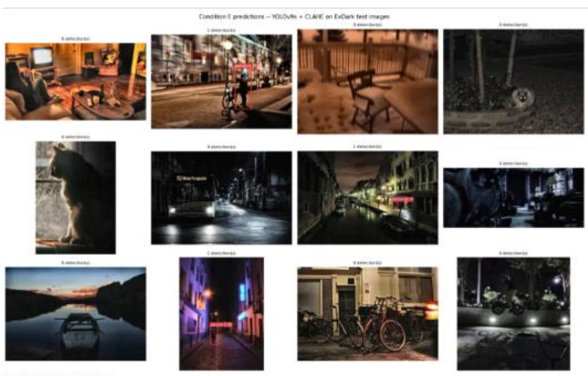

Fig. 7. Condition E (YOLOv9s + CLAHE) predictions on twelve randomly sampled ExDark test images, with detection count and confidence overlaid.

E. Failure Case Analysis

Three recurring failure patterns were identified through manual inspection of the qualitative prediction grid and the lowest-confidence true positives in the test set. The first and most common pattern is small-object failure: consistent with the bounding-box statistics reported during exploratory data analysis, where the median pedestrian box covers only 2.4% of image area and 75% of boxes cover less than 7.9%, the detector's confidence scores degrade sharply as the pedestrian's apparent size shrinks below roughly 3% of frame area, a pattern also reported by Wang et al. [11] for small-object detection in low-light tunnel imagery. The second pattern is silhouette-only failure, occurring in scenes with strong backlighting from a single dominant source (e.g., an oncoming headlight or a doorway light), where the pedestrian appears as a near-uniform dark silhouette with no internal texture; this mirrors the high-dynamic-range clipping challenge described in Section III and is not fully addressed by CLAHE, since CLAHE cannot recover detail that was clipped to pure black during image capture rather than merely compressed into a narrow tonal range. The third pattern is motion-blur failure, most apparent in the augmentation-trained conditions (C, D), where pedestrians in genuine motion (e.g., mid-stride crossing a street) are missed at a higher rate than stationary pedestrians, suggesting that the fixed motion-blur kernel range (3-7 pixels) used in training may not fully span the blur magnitudes present in the real ExDark test images, an interaction between synthetic and real degradation that would benefit from further tuning in future work.

These three failure modes are not mutually exclusive; the most difficult test images typically exhibit at least two simultaneously (e.g., a small, backlit, partially blurred pedestrian at frame edge). This compounding effect helps explain why even the best-performing condition (E) achieves a recall of only 0.475: a substantial fraction of the test set's pedestrian instances fall into one or more of these categories, and no single intervention evaluated in this project, CLAHE, synthetic augmentation, or architecture choice, fully resolves all three simultaneously. This observation directly motivates the multispectral fusion direction proposed in Section V, since silhouette-only and small-object failures are fundamentally information-loss problems in the visible spectrum that infrared sensing can partially address.

F. Inference Efficiency

Since real-time traffic-camera deployment is a stated motivation of this project, inference latency was recorded for every condition alongside detection accuracy. Table IV reports the average per-image inference time (excluding pre/post-processing overhead) measured on the same Kaggle Tesla T4 GPU used for training, along with the corresponding throughput in frames per second.

Cond.	Model	Inference (ms)	Throughput (FPS)
A	YOLOv8s	5.98	167
A2	YOLOv8s+CLAHE	7.18	139
B	YOLOv8s+CLAHE (FT)	7.13	140
C	YOLOv8s+Aug (FT)	7.11	141
D	YOLOv8s+CLAHE+Aug	7.25	138
E	YOLOv9s+CLAHE (FT)	8.75	114

TABLE IV. Measured inference latency and throughput per condition (T4 GPU, batch size 16, 640x640 input).

Two observations follow from Table IV. First, all six conditions sustain real-time-capable throughput well above typical traffic-camera frame rates (commonly 15-30 FPS), meaning that the accuracy gains reported in Table II are not offset by an impractical latency cost; even the slowest condition (E, 114 FPS) leaves substantial headroom for a production deployment pipeline that also performs frame decoding, tracking, or alert logic downstream of detection. Second, CLAHE preprocessing itself contributes a small, consistent latency overhead (roughly 1.1-1.3 ms/image, visible by comparing Condition A to A2), consistent with the 'negligible computational overhead' characterization of CLAHE reported by Tomas et al. [15], since it requires only a lightweight per-tile histogram operation rather than a neural-network forward pass. Third, YOLOv9s (Condition E) is measurably slower than every YOLOv8s condition despite having 36% fewer parameters, which initially appears counterintuitive; this is explained by YOLOv9's GELAN and AConv blocks trading parameter count for additional sequential layers (197 fused layers at inference for YOLOv9s versus 73 for YOLOv8s in our runs), illustrating that parameter count alone is an unreliable proxy for inference latency and that architecture choice for edge deployment should be validated empirically rather than assumed from model size. This latency/accuracy trade-off between Conditions B and E, where E gains 0.006 mAP@0.5:0.95 and 0.045 precision at the cost of roughly 20% lower throughput, is a concrete, deployment-relevant decision point that a practitioner adopting this pipeline would need to weigh against their specific hardware and frame-rate requirements.

G. Reproducibility

All data preparation steps (annotation format conversion, official train/validation/test partitioning, CLAHE parameter selection, and augmentation pipeline construction) were implemented as standalone, independently runnable scripts, decoupling local data preprocessing (performed on CPU) from GPU-dependent fine-tuning (performed on Kaggle T4 instances). This separation was a practical necessity given hardware constraints, but also has a methodological benefit: each preprocessing decision, most notably the CLAHE clip-limit selection and the augmentation parameter tuning, was validated visually against saved before/after image pairs prior to being used in any training run, rather than accepting library defaults uncritically. The complete metrics table (Table II) was accumulated incrementally across sessions into a single CSV artifact, appended to after each condition's evaluation, which both guarded against data loss from Kaggle's session time limits and provided a running, auditable record of every configuration change made during the project.

VII. CONCLUSION AND DISCUSSION

This project implemented and evaluated a complete pipeline for nighttime pedestrian detection on the ExDark dataset, isolating the individual and combined contributions of CLAHE preprocessing, synthetic nighttime augmentation, and architecture choice (YOLOv8s vs. YOLOv9s). Three findings stand out as the project's primary contributions relative to prior work. First, CLAHE's benefit is almost entirely contingent on fine-tuning; applied alone at inference time to a non-adapted model, it produced a statistically negligible improvement (mAP@0.5:0.95 from 0.118 to 0.119), which qualifies and partially contradicts prior reports of meaningful CLAHE-only gains [15] and suggests that CLAHE's effectiveness as a zero-retraining intervention may be architecture- or dataset-dependent. Second, under matched fine-tuning budgets, CLAHE alone outperformed both synthetic nighttime augmentation alone and their combination, indicating that naively stacking preprocessing and augmentation interventions can compound distributional shift rather than complementing it. Third, YOLOv9s achieved the best overall mAP@0.5:0.95 and precision under identical preprocessing and training conditions while using substantially fewer parameters than YOLOv8s, supporting the case for newer architectures in resource-constrained, real-time traffic-camera deployment scenarios.

A secondary contribution is the lighting-condition subset analysis, which revealed that ExDark's ten lighting-condition labels do not form a monotonic darkness gradient as commonly assumed; the 'Low/Ambient/Object' subset outperformed the 'Screen/Window/Shadow/Twilight' subset, the opposite of our initial hypothesis. We report this transparently as a methodological caveat for future researchers using ExDark's category labels as a proxy for scene darkness.

This work has limitations. Training was capped at 50 epochs with early-stopping patience of 15 to balance convergence against the compute budget available on free-tier Kaggle GPU instances; all reported runs converged and stopped between epochs 24 and 33, suggesting the budget was adequate, though a larger compute budget could further validate this. The lighting-subset groupings used category labels rather than measured scene luminance, which our own results show can be misleading; a luminance-based regrouping is a natural extension. Finally, all experiments were confined to a single-class (person) formulation; extending the pipeline to the full 12-class ExDark taxonomy would test whether the reported findings generalize beyond pedestrian detection.

Three directions are proposed for future work. First, fusing visible-spectrum and infrared imagery, following the paired-modality approach of LLVIP [8], could address the fundamental information-loss ceiling that RGB-only sensors face in near-total darkness, which no preprocessing or augmentation strategy can fully overcome. Second, transformer-based detectors (e.g., RT-DETR or DETR-family architectures) could be evaluated under the same CLAHE and augmentation protocol established here, to test whether the CLAHE-alone-outperforms-augmentation finding generalizes beyond CNN-based YOLO architectures. Third, given the precision/recall trade-offs observed across conditions (e.g., Condition E's higher precision but marginally lower recall than Condition B), future work should explore deployment-oriented calibration, adjusting confidence thresholds against real-world cost asymmetries between missed pedestrian detections and false alarms, and validate any resulting model on embedded or edge hardware representative of real traffic-camera deployments. A fourth, complementary direction is temporal smoothing: since traffic cameras produce continuous video rather than independent frames, incorporating a lightweight multi-object tracker (e.g., ByteTrack or a Kalman-filter-based tracker) downstream of per-frame detection could recover some of the recall lost to single-frame failure modes identified in Section IV-E, particularly transient motion-blur and small-object misses, by propagating detections across nearby frames where the pedestrian was more clearly visible.

From an implementation standpoint, this project was built entirely from open-source tooling, Python, OpenCV [20] for the CLAHE module, Albumentations [17] for the augmentation pipeline, and the Ultralytics package [18] for both YOLOv8 and YOLOv9 training and evaluation, with all data preparation performed locally and all GPU-dependent fine-tuning performed on Kaggle's free-tier T4 instances. This choice of stack was deliberate: every component used here is freely available and well-documented, so the complete pipeline, from raw ExDark annotations to the final six-condition comparison in Table II, can be reproduced by another researcher without specialized infrastructure or paid compute, which we view as a meaningful practical contribution in its own right for a benchmark intended to inform real-world traffic-safety deployment decisions.

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